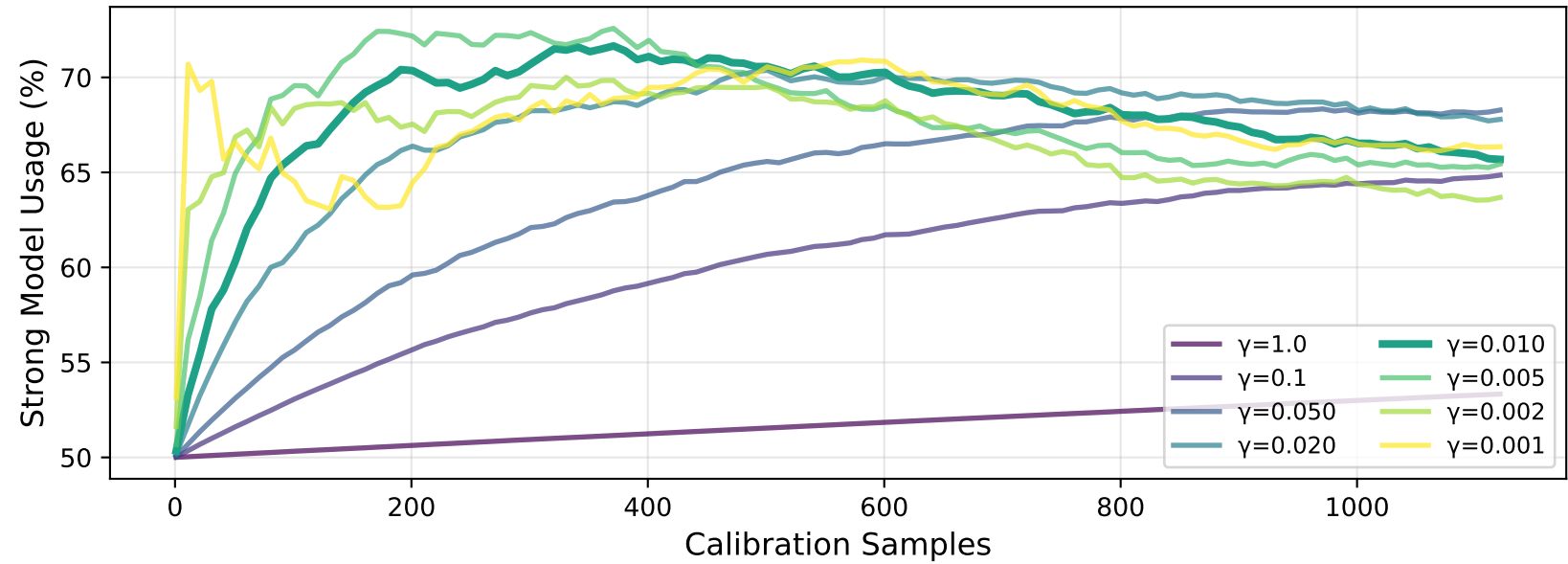
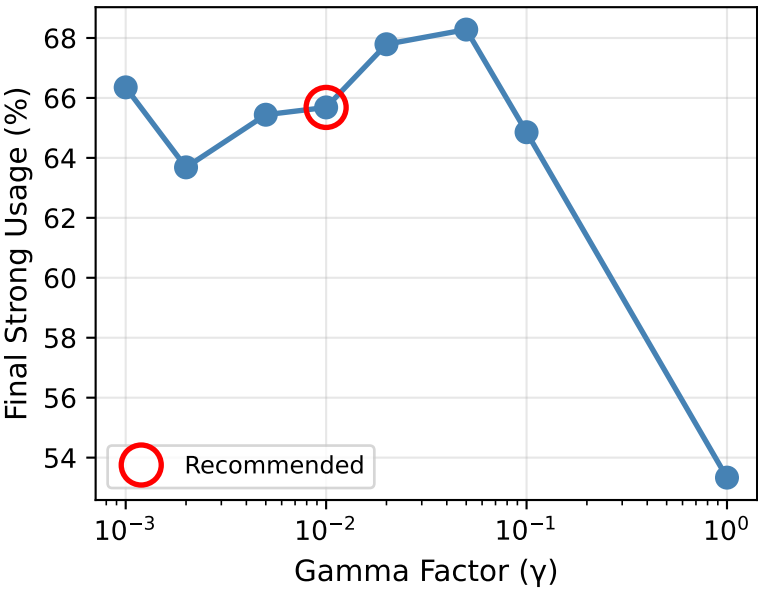


Figure 3: Optimal Gamma Calibration Analysis
Finding the Balance Between Warmup Priors and Calibration Data

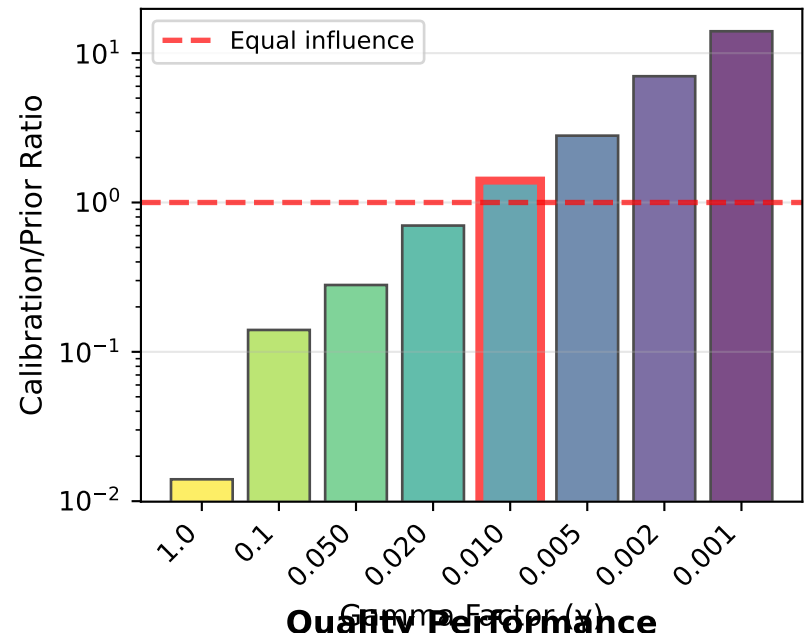
Policy Adaptation Curves by Gamma



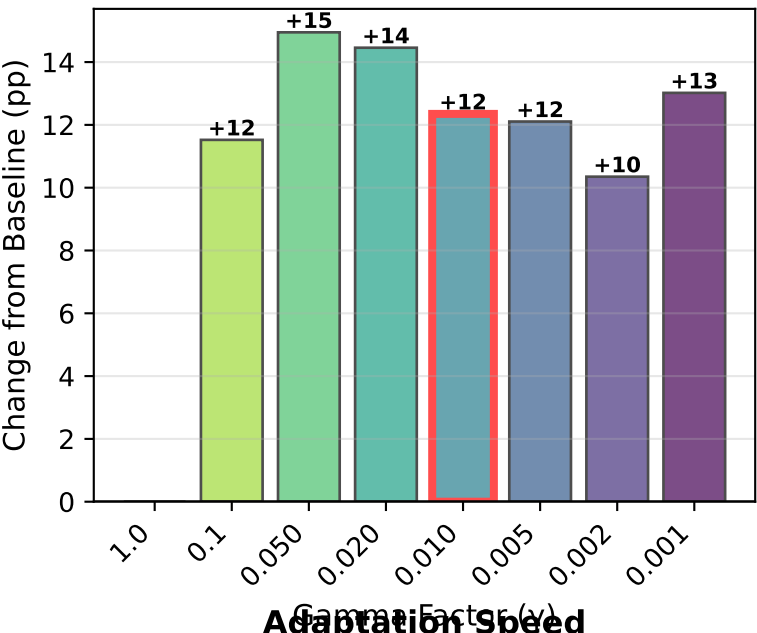
Prior Weakening Effect



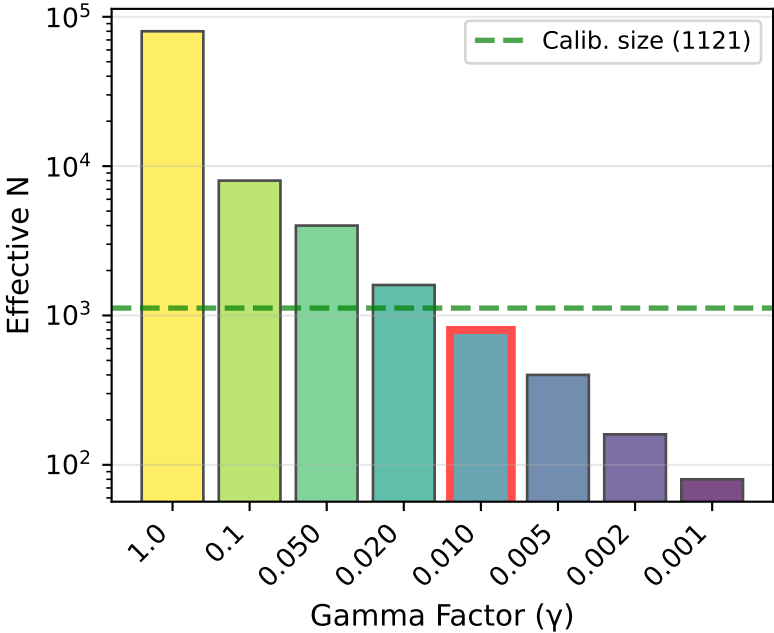
Influence Balance



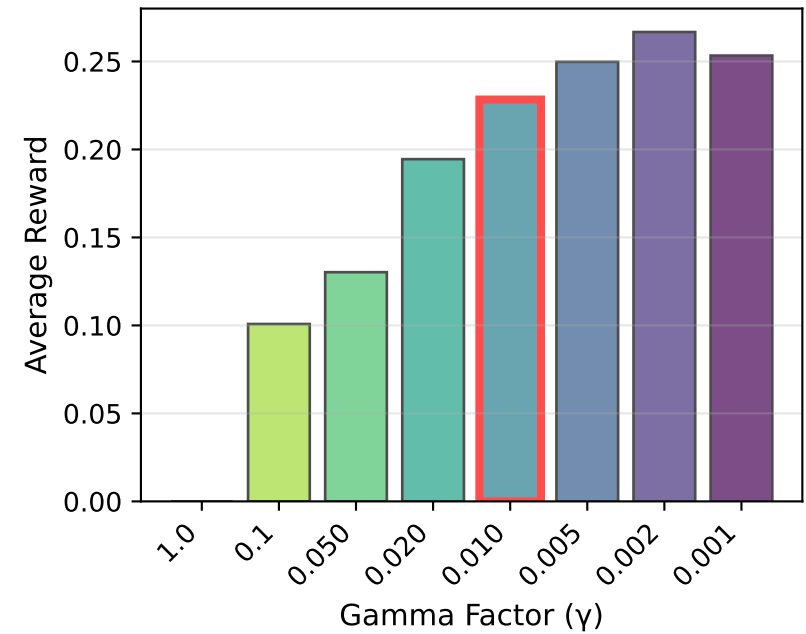
Adaptation Magnitude



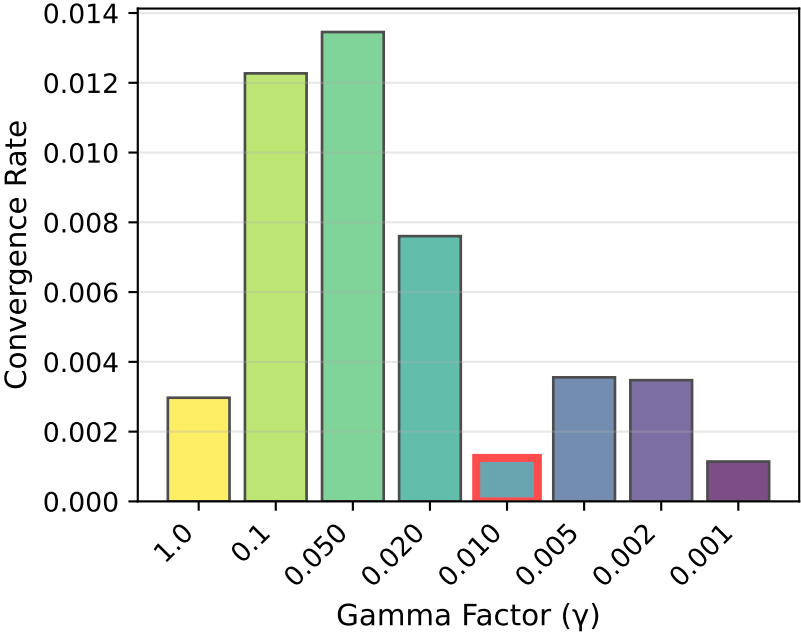
Prior Strength



Quality Performance



Adaptation Speed



OPTIMAL GAMMA ANALYSIS

Dataset:

- Calibration: 1,121 samples
- Warmup: 80,000 samples
- Models: Mixtral 8x7B Instruct vs Gpt 4 Turbo

Recommended: $\gamma = 0.010$

Key Metrics:

- Strong usage: 65.7%
- Avg reward: 0.2284
- Calib/Prior: 1.401
- Eff. N: 800

Baseline ($\gamma=1.0$):

- Strong usage: 53.3%
- Change: +12.4 pp

Next Steps:

1. Use $\gamma=0.010$ for calibration
2. Run `calibrate_router.py`
3. Evaluate on holdout set